Proceedings of the International Conference on Trends in Material Science and Inventive Materials (ICTMIM-2025)
IEEE Xplore Part Number: CFP250J1-ART; ISBN: 979-8-3315-0148-8

2025 5th International Conference on Trends in Material Science and Inventive Materials (ICTMIM) | 979-8-3315-0148-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICTMIM65579.2025.10987928

# IoT and ML for Air Pollution Monitoring with Real Time Data and Prediction of Health Advisory

Dr.M.Janardhan
Associate Professor, Department of Artificial Intelligence, G. Pullaiah College of Engineering and Technology, Kurnool, Andhra Pradesh, India.
m.janardhan0105@gmail.com

Nagella Rakshitha
Student, Department of Internet of Things, G. Pullaiah College of Engineering and Technology, Kurnool, Andhra Pradesh, India.
nrakshi19@gmail.com

Ediga Sravani
Student, Department of Internet of Things, G. Pullaiah College of Engineering and Technology, Kurnool, Andhra Pradesh,India.
sravaniediga175@gmail.com

Dudekula Mubeena Afroz
Student, Department of Internet of Things, G. Pullaiah College of Engineering and Technology, Kurnool, Andhra Pradesh, India.
mubeenadudekula2003@gmail.com

Nelli Supriya
Student, Department of Internet of Things, G. Pullaiah College of Engineering and Technology, Kurnool, Andhra Pradesh, India.
nellisupriya90@gmail.com

***Abstract — Air pollution is a major environmental and health concern, especially in urban areas where high pollution levels can severely impact individuals with respiratory conditions. This paper presents an IoT and ML-based Air Pollution Monitoring System designed to provide real-time air quality monitoring, predictive analytics, and intelligent route optimization. The system utilizes an IoT module installed at traffic signals, equipped with air quality sensors to monitor CO, $CO_2$, and smoke levels. The collected data is displayed on an LCD screen, informing commuters of pollution levels and health advisories. A mobile application integrates real-time pollution data and ML-based predictions to assess outdoor safety, particularly for individuals with respiratory conditions. The system employs a Decision Tree Classifier to generate personalized health advisories based on pollution levels and user-reported symptoms. Additionally, the system offers intelligent route optimization, suggesting alternative routes with lower pollution exposure using real-time sensor data and historical trends. For seamless operation, Firebase is used for real-time data synchronization, ensuring accurate and timely updates. The integration of IoT and ML enables continuous air quality monitoring and proactive health measures. This research aims to enhance public awareness, encourage healthier urban mobility, and support smart city initiatives by providing data-driven insights and reducing pollution exposure risks.***

***Keywords: IoT, Machine Learning, Air Pollution Monitoring, Smart City, Health Advisory System, Intelligent Route Optimization, Real-Time Data Analysis, Environmental Monitoring, Predictive Analytics, Public Health Awareness, Traffic Signal Integration.***

## 1.INTRODUCTION

Air pollution is a major environmental concern, especially in urban areas where emissions from vehicles, industries, and other sources significantly degrade air quality. Traditional air pollution monitoring systems rely on fixed monitoring stations that provide broad, area-wide data instead of localized, real-time insights. These conventional systems lack mobility, predictive analytics, and personalized health recommendations, making it challenging for individuals, particularly those with respiratory conditions, to make informed decisions about outdoor activities.

Existing air pollution tracking systems primarily focus on data collection and reporting rather than offering actionable solutions. They do not provide instant alerts or alternative route suggestions to help commuters avoid high-pollution areas, limiting their practical effectiveness. Furthermore, most traditional models use fixed threshold classifications, which fail to adapt to real-time fluctuations in pollution levels, making them less effective in providing timely and personalized health advisories.

To overcome these limitations, this research presents an IoT and ML-based Air Pollution Monitoring System that integrates real-time air quality monitoring, predictive analytics, and intelligent route optimization. IoT-enabled air quality sensors, installed at traffic signals, continuously monitor key pollutants such as CO, $CO_2$, and smoke. The collected data is displayed on an LCD screen with health advisories, enabling commuters to take immediate preventive actions based on pollution levels. A key innovation of this system is the use of a Decision Tree


Authorized licensed use limited to: Universitas Brawijaya. Downloaded on August 29,2026 at 12:44:16 UTC from IEEE Xplore. Restrictions apply.

Classifier, which processes real-time pollution data and user-reported symptoms to generate personalized health advisories. Additionally, the system incorporates an intelligent route optimization module that suggests alternative travel paths with lower pollution exposure using real-time sensor data and historical pollution patterns. By leveraging Firebase for seamless data synchronization, the system ensures that users receive instant updates on air quality trends and health advisories, contributing to smart city initiatives for improved urban air quality management.

## 2. LITERATURE REVIEW

With the growing awareness of air quality, particularly in cities, numerous IoT-based monitoring systems have been developed. The objective of such systems is to offer cost-effective, real-time, and scalable solutions to monitor the level of environmental pollution.

Arun Raj et al. [1] emphasized the importance of localized air quality monitoring in cities and suggested a system based on sensors for monitoring pollutants. Gupta et al. [2] extended this idea to smart cities by integrating IoT with data analytics to provide dynamic monitoring and improved decision-making in urban planning.

Some of the works were centered on reviewing and evaluating current systems. Priyanka [3] presented a detailed review of air quality monitoring technology, highlighting the shift from conventional static stations to mobile and IoT-based technology. Caubel et al. [4] presented a new black carbon sensor for dense deployment, which improved the accuracy and spatial resolution of pollution information.

Yamunathangam et al. [5] designed an awareness creation system with pollution monitoring, utilizing IoT to alert the users about impending pollution levels. Phala et al. [6] formulated a system with ISO/IEC/IEEE 21451 standards compliance for interoperability and standardization among sensor networks. Desai and Alex [7] suggested a BeagleBone Black-based IoT air quality prediction system, and Dhingara et al. [8] presented IoT-Mobair, a mobile-based monitoring platform showcasing the wearable and portable trend in air quality systems. This is in line with the research on wearable sensors for personal exposure analysis [9], which highlights the importance of individual monitoring in healthcare studies.

Lo Re et al. [10] offered a mobile-based pollution exposure assessment application, making environmental data accessible to the public. Rukmani et al. [11] integrated IoT, image processing, and big data analytics for monitoring industrial pollution, aligning with the increasing requirement of smart processing in intricate surroundings. Monika Singh et al. [12] and Parmar et al. [14] highlighted low-cost, Arduino-based options, best suited for mass deployment in resource-limited areas. Bathiya et al. [13] considered wireless sensor networks for pollution monitoring, while Verma et al. [15] reiterated the potential of IoT to increase granularity and responsiveness in air quality monitoring systems.

Collectively, these studies document the transformation of air pollution measurement from fixed, centralized systems to decentralized, real-time, IoT-enabled networks focused on accuracy, accessibility, and public awareness.

## 3. PROPOSED SYSTEM

The proposed system integrates IoT, AI, and machine learning to provide a real-time air pollution monitoring solution with predictive analytics and intelligent route optimization. Designed for urban environments, the system enhances public health awareness and air quality management.

### *A. Real-Time Air Quality Monitoring*

IoT-based sensors at traffic signals measure pollutants like CO, $CO_2$, and smoke**.** An LCD screen displays pollution levels with health advisories for commuters.

### *B. Mobile Application for Pollution Insights*

A mobile app provides real-time pollution data and ML-based predictions to assess outdoor safety. Users receive personalized health recommendations, especially for individuals with respiratory conditions.

### *C. Intelligent Route Optimization*

The system suggests alternative routes with lower pollution levels based on real-time and historical data. Helps users minimize exposure to harmful pollutants while commuting.

### *D. Predictive Analytics and Health Advisory System*

Machine learning models analyze trends to predict pollution levels and provide proactive health measures. Displays preventive actions, such as mask recommendations, based on pollution severity.

### *E. Real-Time Data Synchronization with Firebase*

Ensures seamless communication between IoT sensors, the LCD display, and the mobile application. Enables instant updates for accurate, real-time pollution tracking.

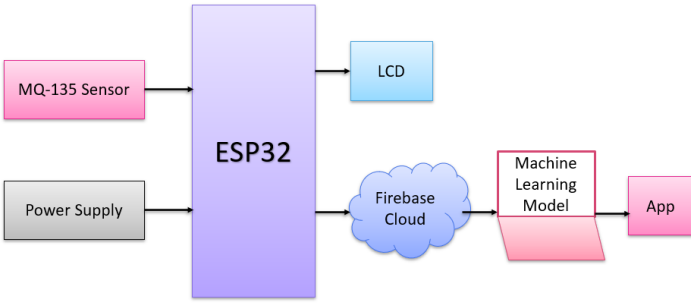

Authorized licensed use limited to: Universitas Brawijaya. Downloaded on August 29,2026 at 12:44:16 UTC from IEEE Xplore. Restrictions apply.

### *F. IoT-Enabled Environmental Monitoring*

Continuously tracks air pollution levels and stores data for long-term analysis. Supports government and environmental agencies in improving air quality regulations.

This integrated solution leverages IoT and ML to empower individuals with data-driven insights, promoting healthier urban navigation and supporting smart city initiatives.

**Block Diagram:**




## 4. HARDWARE COMPONENTS

### *A. ESP32 MicroController*

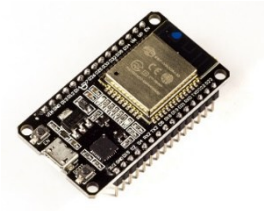


*Fig 1.0: ESP32*

The ESP32 is a versatile microcontroller designed for IoT applications, featuring built-in Wi-Fi and Bluetooth capabilities. It efficiently processes data from various sensors and transmits it to cloud platforms or mobile applications. With support for multiple communication protocols such as I2C, SPI, and UART, the ESP32 seamlessly integrates with environmental monitoring systems. Its low power consumption, high processing speed, and real-time data handling make it an ideal choice for applications like air quality monitoring and smart automation.

### *B. MQ-135*

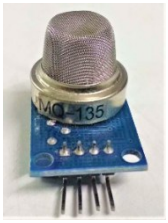



*Fig 1.1: MQ-135 Sensor*

The MQ-135 is an air quality sensor that detects various gases, including $CO_2$, $NH_3$, benzene, and smoke. It works by using a gas-sensitive material that changes resistance when exposed to pollutants. The sensor measures the concentration of gases in the air and provides an analog output that can be processed by microcontrollers like the ESP32. It is widely used in air pollution monitoring systems to assess air quality and detect hazardous gases.

### *C. LCD*



*Fig 1.2: LCD*

An LCD (Liquid Crystal Display) is used to visually present data collected from sensors. It operates by manipulating liquid crystal molecules to control light passage, creating readable text and graphics. In an air pollution monitoring system, the LCD screen displays real-time air pollution level, along with health advisories. LCDs are energy-efficient and offer clear visibility, making them suitable for embedded systems and IoT applications.

## 5. SOFTWARE COMPONENTS

### *A. MU Editor*

Mu Editor is a simple yet powerful Python code editor designed for beginners and developers working with microcontrollers. It provides an intuitive interface with essential features such as syntax highlighting, an interactive REPL, and direct support for MicroPython-based boards like ESP32. Mu Editor allows users to write, test, and upload code efficiently, making it an excellent tool for IoT and embedded system development.


Authorized licensed use limited to: Universitas Brawijaya. Downloaded on August 29,2026 at 12:44:16 UTC from IEEE Xplore. Restrictions apply.

### *B. Firebase*

Firebase is a cloud-based backend platform that provides various services for mobile and web applications. It enables real-time database management, user authentication, cloud storage, and serverless computing. With Firebase, developers can sync data instantly across devices, making it ideal for applications requiring real-time updates. It also offers built-in analytics and AI-powered features to enhance user engagement and app functionality.

### *C. VS Code*

Visual Studio Code (VS Code) is a lightweight yet powerful source code editor designed for multiple programming languages. It offers intelligent code suggestions, debugging tools, and a customizable interface to improve productivity. With an extensive library of extensions, developers can integrate version control, cloud services, and automation tools. Its seamless compatibility with various frameworks makes it a popular choice for software development.

### *D. Android Studio*

Android Studio is the official integrated development environment (IDE) for building Android applications, offering a robust platform with features like a visual layout editor, real-time code analysis, and an advanced emulator for testing. It supports programming in Java and Kotlin, provides built-in debugging tools, and integrates seamlessly with Gradle for efficient project management. With its intelligent code editor and extensive libraries, Android Studio streamlines app development, enabling developers to create responsive and high-performance applications for various Android devices.

### *E. IOT Technology Integration:*

IoT-integrated technology connects physical devices through the internet, enabling real-time data collection, analysis, and automation. Sensors, microcontrollers, and cloud platforms work together to monitor and control environments remotely. This technology enhances efficiency, reduces human intervention, and supports applications in smart cities, healthcare, and industrial automation by enabling seamless communication between devices.

### *F. Machine Learning Technology Integration:*

Machine Learning (ML) enables systems to identify patterns, adapt to data, and improve decision-making without direct programming. By continuously learning from new information, ML enhances automation, optimizes processes, and provides accurate predictions. In this project, a Decision Tree Classifier is utilized to analyze pollution levels and corresponding respiratory symptoms. This model efficiently categorizes pollution levels into predefined ranges and recommends tailored health advisories, such as "Wear a mask" or "Avoid outdoor activities." The integration of the Decision Tree Classifier ensures precise and adaptive responses, leveraging real-time data to provide accurate health recommendations. Furthermore, ML's synergy with IoT and smart applications allows for real-time data analysis, enhanced adaptability, and improved system efficiency, making it a key driver of intelligent technology solutions.



*Fig 2.0: MU Editor*



*Fig 2.1: Android Studio*



*Fig 2.2: Firebase*

## 6. QUANTITATIVE ANALYSIS

A comprehensive quantitative evaluation was conducted to assess the performance of the IoT and ML-based Air Pollution Monitoring System. This evaluation focused on essential performance metrics, including prediction accuracy, response time, system reliability, and overall stability. These metrics are crucial for ensuring accurate health recommendations and real-time air quality monitoring..

The accuracy of pollution level predictions was validated by comparing the system's output with reference data obtained from calibrated sensors and environmental datasets. To categorize pollution levels into "Safe," "Moderate," and "Hazardous," the Decision Tree Classifier was utilized. This model was chosen for its clarity, efficiency in multi-class classification tasks, and ability to provide easily interpretable thresholds for decision-making. Its performance was measured using key metrics such as precision, recall, F1-score, and confusion matrix analysis. The system's response time was also assessed, focusing on how quickly IoT sensors collected data, synchronized it with Firebase, and displayed the results on the


Authorized licensed use limited to: Universitas Brawijaya. Downloaded on August 29,2026 at 12:44:16 UTC from IEEE Xplore. Restrictions apply.

LCD screen and mobile application. The system's ability to deliver timely health advisories based on real-time data was also examined to determine its practicality in urban environments with high traffic density.

The reliability of the system was tested across diverse scenarios, such as sudden spikes in pollution levels and changes in weather conditions. The MQ135 sensor was specifically evaluated for its consistency and accuracy in detecting and processing air quality data. Furthermore, the Firebase platform's capacity to ensure seamless data synchronization and continuous accessibility for mobile users was scrutinized.

The findings demonstrate that the system is capable of reliably monitoring air quality, predicting pollution trends, and providing prompt health advisories. By combining IoT technology for real-time data acquisition, Decision Tree Classifier models for predictive analytics, and cloud-based platforms for efficient data management, the system represents a scalable and robust solution for urban air quality management. The integration of machine learning ensures tailored health recommendations, fostering a proactive approach to mitigating the health impacts of air pollution.

## 7. RESULTS

The IoT and ML-based Air Pollution Monitoring System provides real-time monitoring of air quality through a combination of sensors and an ESP32 microcontroller. The air pollution levels, measured by the system, are displayed on an LCD screen, as shown in Figure 1.6. This module, installed at key locations such as traffic signals, ensures that individuals nearby are immediately informed about the air quality. For example, when the air quality index (AQI) is measured at 111, the system classifies it as "Good," providing users with a clear and concise assessment of environmental conditions.




***Fig 3.0: Display System***

When a user opens the application, they are presented with the current air quality levels, as shown in Figure 1.7, with data retrieved from Firebase and analyzed by the ML model. The application displays the overall air pollution level and offers health advisories based on the predicted pollution index and user-provided symptoms, ensuring personalized recommendations for public safety.




***Fig 3.1:Pollution level***

The health advisory feature allows users to select their respiratory symptoms, such as cough or asthma. Based on these inputs and the pollution levels, the system predicts appropriate health advisories, as shown in Figure 1.8. This feature ensures that users, especially those with respiratory issues, receive tailored preventive measures like "Wear a mask" or "Avoid outdoor activities."




***Fig 3.2:Health Advisory***

When users enter their current location and destination, the system analyzes pollution levels along different routes and suggests alternative paths that minimize exposure to highly polluted areas. By integrating mapping technology, the application visually represents these routes using color codes to indicate varying pollution levels. This feature empowers users to make healthier travel choices by avoiding areas with poor air quality.


Authorized licensed use limited to: Universitas Brawijaya. Downloaded on August 29,2026 at 12:44:16 UTC from IEEE Xplore. Restrictions apply.

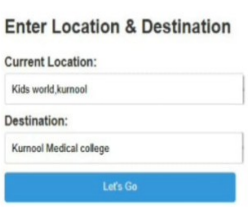



*Fig 3.3:Route selection Interface*



*Fig 3.4:Alternate Routes*

The system's ability to integrate real-time data collection, user inputs, and machine learning predictions ensures scalability and efficiency. It provides valuable insights to urban residents, promoting awareness and empowering them to take action against air pollution. By delivering accurate health advisories and optimized travel routes, the IoT and ML-based Air Pollution Monitoring System acts as a reliable companion for healthy urban living.

## 8. CONCLUSION

The IoT and ML-based Air Pollution Monitoring System provides an effective solution for real-time air quality monitoring, health advisories, and pollution-aware route planning. By integrating IoT sensors, Firebase for real-time data synchronization, and a Decision Tree Classifier for health advisory predictions, the system offers personalized recommendations to users. Air quality information is displayed on an LCD screen near traffic signals, while the mobile app provides real-time pollution updates and alternative travel routes to minimize exposure to poor air quality. However, certain challenges remain, such as sensor accuracy variations due to environmental factors, network reliability for real-time updates, and the need for continuous model optimization as pollution patterns change. Future improvements could include more advanced ML models, sensor calibration refinements, and integration with government air quality databases for broader coverage. Additionally, expanding pollutant detection to include PM2.5 and PM10 could further improve precision. With its scalability, accessibility, and smart city applications, this system can significantly enhance urban air quality management, public health, and sustainable mobility.

## 8. REFERENCES

[1] Arun Raj V., Priya R.M.P., and Meenakshi, V., "Air Pollution Monitoring In Urban Area," International Journal of Electronics and Communication Engineering, 2017.

[2] Harsh Gupta, Dhananjay Bhardwaj, Himanshu Agrawal, Vinay Anand Tikkiwal, Arun Kumar, (2019) 'An IoT Based Air Pollution Monitoring System for Smart Cities', ICSETS.

[3] Priyanka, V., "Review: Air Quality Monitoring System," International Journal of Advanced Research in Computer and Communication Engineering, 5(6), 2016.

[4] J. J. Caubel, T.E. Cados, T.W. Krichstetter, (2018) 'A New Black Crabon Sensor for Dense Air Quality Monitoring Networks', IEEE

[5] Yamunathangam, K. Pritheka, P. Varuna, (2018) 'IoT Enabled Air Pollution Monitoring and Awareness Creation System', International Journal of Recent Technology and Engineering, IRJET.

[6] K. S. E. Phala, A. Kumar, and Gerhard P. Hancke, 'Air Quality Monitoring System Based on ISO/IEC/IEEE 21451 Standards', IEEE.

[7] Nitin Sadashiv Desai, John Sahaya Rani Alex, (2017) 'IoT based air pollution monitoring and predictor system on Beagle Bone Black', International Conference on Nextgen Electronic Technologies, ICNET.

[8] Swati Dhingara, Rajasekhara Babu Madda, Amir H. Gandomi, Rizwan Patan, Mahmoud Daneshmand, (2019) 'Internet of Things Mobile – Air Pollution Monitoring System (IoT-Mobair)', IEEE.

[9] "Wearable Sensors for Analyzing Personal Exposure to Air Pollution," IEEE, 2017.

[10] G. Lo Re, D. Peri, S. D. Vassallo, (2013) 'A mobile application for assessment of air pollution exposure', IEEE

[11] Rukmani, P., Teja, G.K. and Vinay, M.S., 2018. Industrial Monitoring Using Image Processing, IoT and Analyzing the Sensor Values Using Big Data. Procedia Computer Science, 133, pp.991-997.

[12] Monika Singh, Misha Kumari, Pradeep Kumar Chauhan, (2019) 'IoT Based Air Pollution Monitoring System using Arduino', International Research Journal of Engineering and Technology, IRJET.

[13] Bhavika Bathiya, Sanjay Srivastava, Biswajit Mishra, "Air pollution monitoring using wireless sensor network", 2017 IEEE International WIE Conference on Electrical and Computer Engineering , pp. 117-121.

[14] Gagan parmar, sagar lakhani,manju k chattopadhyay, "An IOT based low cost air pollution monitoring system", IEEE 2017 International Conference on Recent innovations in signal processing and embedded systems(RISE)

[15] D. Verma, S. Dhul, R.saini, R.B. Dubey, "IoT Based Air Pollution Monitoring System", International Journal of Innovative Research in Engineering & Management (IJIREM), May 2018


Authorized licensed use limited to: Universitas Brawijaya. Downloaded on August 29,2026 at 12:44:16 UTC from IEEE Xplore. Restrictions apply.